Office of National Academic Test Program (NAT)
สำนักงานโครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.)

# การแข่งขันสอบวัดทักษะวิชาการระดับชาติ (สวช.) ชิงถ้วยพระราชทาน สมเด็จพระกนิษฐาธิราชเจ้า กรมสมเด็จพระเทพรัตนราชสุดาฯ สยามบรมราชกุมารี ประจำปี พ.ศ. 2567 (รอบสุดยอดอัจฉริยภาพระดับประเทศ)

## วิชาภาษาอังกฤษ ระดับชั้นมัธยมศึกษาตอนปลาย

รหัสวิชา อ405 ภาษาอังกฤษ

**รหัสชุดข้อสอบ 405011**

ชื่อ.....นามสกุล.....รหัสวิชา.....

ระดับชั้น.....ห้องสอบ.....วันที่สอบ.....สถานที่สอบ.....

**คำชี้แจงการสอบ**

| | | | |
|---|---|---|---|
| รายละเอียดข้อสอบ | **ข้อสอบฉบับนี้มี 9 หน้า จำนวน 50 ข้อ ระยะเวลา 90 นาที** | | |
| วิธีการตอบ | **ให้ใช้ดินสอ 2B ระบายในวงกลมที่เป็นคำตอบในกระดาษคำตอบ หรือ คลิกเลือกตัวเลือกและยืนยันคำตอบในระบบกระดาษคำตอบออนไลน์** | | |
| เกณฑ์การให้คะแนน | **ส่วน Expression** | **ข้อ 1-10** | **ข้อละ 1.6 คะแนน** |
| **(คะแนนเต็ม 100 คะแนน)** | **ส่วน Reading Comprehension** | **ข้อ 11-25** | **ข้อละ 2.4 คะแนน** |
| | **ส่วน Structure** | **ข้อ 26-40** | **ข้อละ 2.2 คะแนน** |
| | **ส่วน Vocabulary** | **ข้อ 41-50** | **ข้อละ 1.5 คะแนน** |

**ข้อปฏิบัติในการสอบ**

1. ให้ผู้เข้าสอบปฏิบัติตามระเบียบสำนักงานโครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.) ภาคีเครือข่ายโรงเรียนทั่วประเทศ และสถาบันอีซี่ลิชประเทศไทย ว่าด้วยแนวทางปฏิบัติเกี่ยวกับการดำเนินการทดสอบ พ.ศ. 2560 อย่างเคร่งครัด
2. สำหรับสนามสอบปิด ให้ผู้เข้าสอบระบายคำตอบลงในกระดาษคำตอบด้วยดินสอ 2B อย่างถูกต้องและชัดเจน
3. สำหรับสนามสอบออนไลน์ เมื่อผู้เข้าสอบนำโทรศัพท์มือถือหรืออุปกรณ์อิเล็กทรอนิกส์มาสแกน QR Code เพื่อเข้าสู่แบบฟอร์มกระดาษคำตอบ กรอกหรือเลือกคำตอบให้ถูกต้องและครบถ้วน โดยเปิดกล้อง ZOOM ยืนยันตัวตนไว้จนกว่าจะยืนยันกระดาษคำตอบเสร็จ และบันทึกภาพหน้าจอเพื่อเป็นหลักฐาน หลังกดส่งยืนยันคำตอบและปิดกล้อง

**หากผู้เข้าสอบฝ่าฝืนข้อปฏิบัติโครงการ สวช. หรือมีการทุจริต จะมีการดำเนินการ ดังนี้**

1. ไม่ประกาศผลการสอบจากการทดสอบดังกล่าว
2. แจ้งไปยังโรงเรียนต้นสังกัดของนักเรียน ผู้ดูแลของผู้สอบเพื่อดำเนินการตักเตือนและกล่าวโทษ
3. แจ้งไปยังผู้บริหารสถานศึกษาที่เป็นภาคีเครือข่าย สวช. เพื่อพิจารณารับเข้าศึกษา

เอกสารนี้เป็นลิขสิทธิ์ของสำนักงานโครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.)
มูลนิธิโครงการสอบวัดทักษะวิชาการและโอลิมปิกนานาชาติ
**การทำซ้ำหรือดัดแปลงหรือเผยแพร่เอกสารฉบับนี้จะถูกดำเนินคดีตามกฎหมาย**

**Part I: Expression (Items 1 – 10)**
**Choose the best answer.**

**Situation: Three friends are discussing about automotive engineering**

Ryan: The innovations in automotive engineering are remarkable. Have you read about how regenerative braking systems are _____(1)_____?

Sophia: Yes, they're fascinating. The ability to recuperate kinetic energy and convert it into electrical energy is a paradigm shift in vehicle technology.

Liam: True, but I think the integration of artificial intelligence in autonomous vehicles is even more transformative. It's _____(2)_____ of computational mechanics.

Ryan: Absolutely. Machine learning algorithms are enabling cars to make predictive decisions, but the reliability of such systems is still contentious.

Sophia: That's a fair point. The ethical dilemmas _____(3)_____ in self-driving cars add another layer of complexity to their implementation.

Liam: Not to mention, the structural integrity of lightweight materials like carbon fiber composites is still under scrutiny for mass production.

Ryan: Agreed. While they improve fuel efficiency, scaling their use without compromising safety remains a significant challenge.

Sophia: Nonetheless, the progress in battery technology, particularly with solid-state batteries, might mitigate some of these challenges.

Liam: Absolutely. Automotive engineering is_____ (4)_____of a new era, blending sustainability with _____(5)_____.

**1.**
1. revolutionized energy efficiency
2. revolutionizing energy efficiency
3. revolutionized energy efficiently
4. energizing revolutionary efficiently
5. energized revolutionary efficiently

**2.**
1. pushing the boundaries
2. making the boundaries
3. penetrating the boundaries
4. creating the walls
5. climbing the walls

**3.**
1. surround decision-making
2. surrounding decision-making
3. surrounded by decisions
4. making decision surrounding
5. making decision surrounded

**4.**
1. on the horizon
2. on top
3. on the top
4. on the cusp
5. on the side

**5.**
1. cutting-edge technology
2. cutting technology
3. edgeless technology
4. double-edged sword technology
5. cutting sword technology

**Situation: Teacher and students are discussing about Physics.**

Teacher: _____(6)_____ the concept of gravitational waves? They're a prime example of how Einstein's theory of general relativity continues to _____(7)_____.

Student: Yes, I've read about them. The idea that spacetime itself ripples due to massive objects colliding is fascinating. LIGO's detection in 2015 was a real game changer.


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Teacher: Exactly. It confirmed predictions made a century ago. But don't forget how this ties into spacetime curvature—it's the crux of the theory.

Student: That's _____(8)_____. How does the curvature of spacetime dictate the motion of celestial bodies? It's not as clear as day to me.

Teacher: Think of it this way: mass distorts spacetime like a heavy ball on a stretched sheet, _____(9)_____ along the curvature.

Student: Oh, so the planets orbit the sun because of this distortion, not because of a force pulling them directly.

Teacher: Precisely. It's a departure from Newton's perspective of gravity as a force. Einstein's view reshaped our understanding of the cosmos.

Student: And does this apply to black holes too? Their gravitational pull seems so extreme—it's almost like they _____(10)_____.

Teacher: In a way, yes. Black holes create singularities, where spacetime curvature becomes infinite. They're fascinating anomalies in physics.

Student: It's incredible how physics continues to evolve. Every discovery feels like we're just about to unveil the universe's mysteries.

**6.** 1. Did you read 2. Did you review
3. Have you noticed 4. Have you presented
5. Have you reviewed

**7.** 1. stand the test of time 2. stand out through time
3. stand up of time 4. test the time stand
5. test time

**8.** 1. what I struggle 2. why I'm struggling for
3. why I'm struggling with 4. why I struggle
5. what I'm struggling with

**9.** 1. caused by smaller moving objects 2. caused by smaller objects to move
3. causing smaller objects to move 4. causing smaller moving objects
5. causes smaller objects to move

**10.** 1. set the rules 2. follow the rules
3. break the rules 4. create the rules
5. bend the rules

**Part II: Reading Comprehension (Items 11 – 25)**

| **The Psychology of Procrastination** |
|---|
| Procrastination is a common phenomenon that affects individuals across various professions and age groups. Psychologists define procrastination as the voluntary delay of an intended action despite expecting to be worse off for the delay. It is often characterized by an irrational gap between intention and action. Despite its prevalence, the reasons behind procrastination remain complex and multifaceted. |

Research suggests that procrastination is not simply a matter of poor time management but is deeply linked to emotional regulation. People often procrastinate because they want to avoid unpleasant tasks or because they fear failure. This is compounded by the human brain's preference for immediate rewards over delayed gratification, a concept known as temporal discounting. For example, watching a TV show may seem more appealing in the moment than preparing for a presentation due next week.

Interestingly, recent studies have revealed a connection between procrastination and perfectionism. Individuals who set excessively high standards for themselves may delay tasks out of fear of not meeting their own expectations. This fear creates a paradox where the procrastinator becomes trapped in a cycle of delay and self-criticism.

Technological advancements have also exacerbated procrastination. The constant influx of notifications, social media updates, and online entertainment provides endless distractions. These distractions are engineered to capture attention, making it increasingly challenging to focus on demanding tasks.

Overcoming procrastination requires a combination of strategies. Cognitive-behavioral techniques, such as breaking tasks into smaller, manageable parts, can help individuals reduce the perceived difficulty of a task. Additionally, practicing self-compassion instead of harsh self-criticism has been shown to reduce procrastination by alleviating feelings of guilt and shame.

In summary, procrastination is a multifaceted behavior rooted in emotional, psychological, and technological factors. While it may seem like a minor habit, its implications for productivity and mental well-being are significant.

**11.** According to the passage, how do psychologists define procrastination?

1. An unavoidable delay in completing tasks.
2. The preference for immediate rewards over future benefits.
3. The voluntary delay of an action despite knowing the consequences.
4. A sign of poor time management skills.
5. An irrational gap between intention and action.

**12.** What is the primary reason people procrastinate, as per the passage?

1. Lack of planning and organization
2. Emotional regulation difficulties
3. Overconfidence in their abilities
4. The influence of social media
5. Excessively high standards for themselves.

**13.** What does the term "temporal discounting" mean in the context of the passage?

1. The ability to manage time effectively.
2. The habit of prioritizing difficult tasks over easy ones.
3. The technique of delaying tasks intentionally.
4. The personality that prefers satisfaction,
5. The preference for immediate rewards over delayed ones.

**14.** What paradox is described regarding procrastination and perfectionism?

1. Procrastinators delay tasks but still meet high standards.
2. Fear of failure leads to delay, which causes further self-criticism.
3. Perfectionists complete tasks quickly to avoid failure.
4. Procrastination has no connection with perfectionism.
5. Perfectionism only focuses on rewards.

**15.** Which strategy is NOT mentioned as a way to overcome procrastination?

1. Breaking tasks into smaller parts.
2. Seeking help from a professional therapist.
3. Practicing self-compassion.
4. Reducing feelings of guilt and shame.
5. All are mentioned.

**Renewable Energy – The Future of Power**

As the global demand for energy continues to rise, the reliance on fossil fuels poses significant challenges, including environmental degradation and resource depletion. Renewable energy sources such as solar, wind, hydroelectric, and geothermal power are increasingly being recognized as viable alternatives. They are not only sustainable but also help mitigate the adverse effects of climate change.

Solar power, harnessed through photovoltaic cells, has become one of the fastest-growing renewable energy sources. Countries with abundant sunlight, such as India and Australia, have invested heavily in solar farms. Similarly, wind power has gained momentum, especially in regions with strong and consistent winds like Northern Europe and the American Midwest. Offshore wind farms, though expensive, have proven to be more efficient due to the absence of physical obstructions.

Hydroelectric power, generated by the movement of water, is one of the oldest forms of renewable energy. Large-scale projects such as dams can produce substantial amounts of electricity. However, they can also disrupt local ecosystems and displace communities, raising ethical concerns. Geothermal energy, on the other hand, uses heat from the Earth's core. While its availability is geographically limited, it provides a reliable and consistent power supply.

Despite these advancements, the transition to renewable energy faces several obstacles. High initial investment costs, the intermittency of certain energy sources, and the need for advanced storage technologies are significant barriers. Governments play a crucial role in addressing these challenges by offering subsidies, investing in research, and encouraging private sector participation.

In conclusion, while renewable energy holds immense promise, realizing its full potential requires overcoming technical, economic, and social hurdles. As the world moves toward a greener future, the adoption of renewables is not just a choice but a necessity.

**16.** According to the passage, what is one of the main benefits of renewable energy?

1. It eliminates energy storage challenges.
2. It helps mitigate climate change.
3. It is less expensive than fossil fuels.
4. It degrades environment.
5. It completely replaces the need for fossil fuels.

**17.** Why are offshore wind farms considered more efficient?

1. They use less expensive technology.
2. They are easier to maintain.
3. They cause less environmental impact.
4. They are the oldest forms of energy.
5. They produce more power due to stronger winds.

**18.** What is one disadvantage of hydroelectric power mentioned in the passage?

1. It produces less electricity than solar power.
2. It cannot be used in modern infrastructures.
3. It can harm ecosystems and displace communities.
4. It is more expensive than geothermal energy.
5. It is a sustainable project.

**19.** Why is geothermal energy not widely used?

1. It is too expensive to harness.
2. It requires constant government subsidies.
3. It provides inconsistent power supply.
4. Its availability is limited to specific locations.
5. It produces less energy than other renewable sources.

**20.** What is a major barrier to the adoption of renewable energy?

1. Lack of public awareness.
2. Limited availability of wind and sunlight.
3. High initial investment costs.
4. It doesn't encourage private sectors.
5. Difficulty in obtaining government approval.

**21.** What role do governments play in promoting renewable energy, according to the passage?

1. They regulate energy usage to prevent overconsumption.
2. They provide funding and encourage private investment.
3. They prioritize fossil fuels to maintain economic stability.
4. They standardize energy storage technologies globally.
5. They displace communities.

**Artificial Intelligence and the Future of Work**

Artificial Intelligence (AI) is rapidly transforming the global workforce. From automating repetitive tasks to making complex decisions, AI is reshaping industries and redefining the nature of employment. While this technological advancement offers numerous benefits, it also raises critical questions about job security and the future role of human workers.

AI has proven to be exceptionally effective in sectors such as healthcare, finance, and logistics. In healthcare, AI-powered diagnostic tools can analyze medical data with unprecedented speed and accuracy, assisting doctors in identifying diseases. In finance, algorithms are used to detect fraudulent transactions and predict market trends. Logistics companies employ AI systems to optimize supply chains and improve delivery efficiency.

However, the rise of AI also poses significant challenges. Automation is expected to replace millions of jobs, particularly in manufacturing and retail sectors. Critics argue that while AI creates new opportunities, the jobs it generates often require specialized skills, leaving less-skilled workers at a disadvantage. This has led to a growing demand for reskilling programs to prepare workers for an AI-driven economy.

Another pressing issue is the ethical implications of AI. Decisions made by AI systems can lack transparency, making it difficult to understand how certain outcomes are reached. This phenomenon, known as the "black box" problem, raises concerns about accountability, particularly in high-stakes environments such as law enforcement and healthcare.

Despite these challenges, experts remain optimistic about the potential of AI. By complementing human capabilities rather than replacing them, AI can lead to greater innovation and productivity. Governments and organizations must work together to establish policies that ensure equitable access to AI technologies and address the societal impact of automation.

In conclusion, while AI is undoubtedly a game-changer for the workforce, its integration must be managed carefully to maximize benefits and minimize risks.

**22.** What tone does the author adopt toward AI in the passage?

1. Pessimistic and critical.
2. Cautiously optimistic.
3. Entirely dismissive.
4. Neutral and indifferent.
5. Pessimistically bias

**23.** What is the "black box" problem mentioned in the passage?
1. AI systems that fail to work in high-stakes environments.
2. The lack of transparency in AI decision-making processes.
3. The inability of AI to handle complex ethical dilemmas.
4. The increasing costs of developing AI technologies.
5. It doesn't provide an accurate account.

**24.** How can AI complement human capabilities?
1. By eliminating all manual tasks in the workplace
2. By enhancing innovation and productivity
3. By creating specialized jobs for all workers
4. By reducing the need for government policies
5. By creating an automative society

**25.** What do experts suggest to address the societal impact of AI?
1. Limiting AI development to certain industries.
2. Creating reskilling programs and equitable policies.
3. Prioritizing automation over human labor.
4. Restricting the use of AI in high-stakes environments.
5. Focusing on the impact of all stakeholders.

**Part III: Structure and Writing (Items 26 – 40)**

**26.** Although the scientist's theory was initially dismissed by her peers, it later gained widespread acceptance after _____.
1. it was proven to be flawed
2. substantial evidence supported her claims
3. the community decided to ignore it
4. the researchers decided to replicate it
5. other researchers failed to replicate her findings

**27.** The environmental activist argued that deforestation, _____ the loss of biodiversity, also contributes significantly to climate change.
1. which leads to
2. despite leading to
3. that leads to
4. as it leads to
5. in which will lead to

**28.** By the time the rescue team arrived, the hikers, who _____ for hours in the freezing cold, were too weak to walk on their own.
1. waited
2. were waiting
3. had been waiting
4. would wait
5. has been waiting

**29.** The politician's speech was criticized because, while it addressed the economic crisis, it failed to propose _____ that could effectively resolve the issue.
1. any viable solutions
2. many superficial ideas
3. several critical arguments
4. a strong opposition
5. a pessimistic support

**30.** The book, which has been hailed as a literary masterpiece, explores themes of identity and belonging, _____ through the lens of historical events.
1. portraying them vividly
2. while ignoring them entirely
3. but fails to connect them
4. despite being disconnected
5. disconnecting them

**31.** Despite the complexity of the experiment, the students managed to complete it successfully, _____ the professor's detailed instructions.
1. due to
2. in spite of
3. regardless of
4. along with
5. in regards to

**32.** Many employees expressed frustration over the company's new policies, which, rather than improving efficiency, _____.
1. helped to streamline processes
2. made their work significantly harder
3. were praised by external reviewers
4. resolved long-standing issues
5. offered them a flexible workflow

**33.** The athlete continued to train rigorously every day, even though _____.
1. his skills showed significant improvement
2. he was already at the peak of his performance
3. he had no intention of competing again
4. he was widely regarded as the best
5. his coach was evaluated him the top

**34.** Not until the meeting was over_____how important the discussion had been.
1. the participants realized
2. the participants had realized
3. did realize the participants
4. had realized the participants
5. did the participants realize

**35.** Under no circumstances_____to disclose the confidential information to anyone.
1. are you allowed
2. you are allowed
3. you will allow
4. will you allow
5. will allow

**Directions: Choose the answer which is grammatically <u>incorrect</u>.**

**36.** **(1)** **<u>Neither</u>** the teacher nor the students **(2)** **<u>was</u>** aware that the assignment, **(3)** **<u>which</u>** **(4)** **<u>had been distributed</u>** last week, **(5)** **<u>were missing</u>** several pages.

**37.** **(1)** **<u>If</u>** the researchers **(2)** **<u>would have conducted</u>** more **(3)** **<u>thorough</u>** experiments, they might have discovered a **(4)** **<u>groundbreaking</u>** result **(5)** **<u>earlier</u>**.

**38.** Trump on Wednesday urged the US Supreme Court **(1)** **<u>to pause</u>** his **(2)** **<u>sentencing</u>** in the hush money case set **(3)** **<u>for</u>** Friday, a **(4)** **<u>high</u>** unusual request that relies in part on the court's decision last year **(5)** **<u>to grant</u>** him broad immunity from criminal prosecution.

**39.** **(1)** **<u>Among</u>** those detained **(2)** **<u>was</u>** the head of the hospital, Dr. Hussam Abu Safiya, **(3)** **<u>whom</u>** **(4)** **<u>Israeli</u>** authorities suspect of being "a Hamas terrorist operative" – an accusation **(5)** **<u>in which</u>** authorities have provided no evidence.

**40.** Meta's **(1)** **<u>newly</u>** appointed Chief of Global Affairs Joel Kaplan told Fox on Tuesday that Meta's partnerships with **(2)** **<u>three-party</u>** fact checkers were well **(3)** **<u>intentioned</u>** at the outset but there's just been **(4)** **<u>too much</u>** political bias in what they choose to fact check and **(5)** **<u>how</u>**.

**Part IV: Vocabulary (Items 41 – 50)**

**Complete the sentence with the appropriate vocabulary**

**41.** The government implemented stringent measures to_____the spread of the contagious virus.
1. exacerbate 2. mitigate 3. amplify 4. neglect 5. integrate

**42.** Despite the adversity she faced, her_____determination enabled her to achieve success against all odds.
1. unwavering 2. erratic 3. superficial 4. apathetic 5. equitable

**43.** The company's decision to invest in renewable energy projects was seen as a_____move toward sustainability.
1. regressive 2. progressive 3. detrimental 4. counterproductive 5. contextual

**44.** The teacher commended the students for her_____analysis of the complex novel, highlighting her depth of understanding.
1. superficial 2. profound 3. cursory 4. shallow 5. credible

**45.** The athlete's_____performance at the championship earned her a standing ovation from the crowd.
1. mediocre 2. exemplary 3. subpar 4. forgettable 5. speculative

**46.** The charity received a_____donation from the philanthropist, allowing them to fund several new projects.
1.meager 2. substantial 3. insubstantial 4. negligible 5. discordant

**47.** The professor's lecture was filled with such_____terminology that many students found it difficult to follow without prior preparation.
1. esoteric 2. transparent 3. rudimentary 4. predictable 5. repudiation

**48.** In a bid to increase transparency, the government passed legislation requiring corporations to disclose their financial practices, a move that was seen as_____by critics.
1. perfunctory 2. revolutionary 3. clandestine 4. inadvertence 5. incoherence

**49.** While the film received critical acclaim for its cinematography, its convoluted plot left many viewers feeling_____rather than entertained.
1. enlightened 2. perplexed 3. indifferent 4. gratified 5. empirical

**50.** Despite the increasing competition in the tech industry, the company managed to_____ its dominance through consistent innovation and strategic partnerships.
1. relinquish 2. perpetuate 3. diminish 4. disregard 5. laud